

FINTRACK – PERSONAL&EXPENSE MANAGEMENT SYSTEM

A PROJECT REPORT

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ABSTRACT

FinTrack: Family and Expense Management System is a smart web-based application developed to help users efficiently manage personal and family expenses. The system enables users to record daily expenses, monitor spending patterns and manage budgets effectively. The application provides separate modules for personal and family expense management. Users can add, edit and delete expenses, set monthly budgets, create category-wise spending limits and receive alerts when spending exceeds predefined limits.

A unique feature of the system is family expense management with shared expense splitting, where expenses can be divided among multiple family members automatically. The system also provides visual analysis through charts and monthly comparisons to help users understand spending behavior. Additionally, the system generates professional expense reports in PDF format and supports calendar-based expense tracking.

The application works offline using a local Flask server and SQLite database, making it simple, efficient and user-friendly. The project is developed using Python Flask, SQLite, HTML, CSS and JavaScript to provide an efficient financial tracking solution for individuals and families.

LIST OF ABBREVIATIONS

ABBREVIATIONS	EXPANSION
AI	Artificial Intelligence
API	Application Programming Interface
CSS	Cascading Style Sheets
HTML	Hyper Text Markup Language
IDE	Integrated Development Environment
SQL	Structured Query Language
PDF	Portable Document Format
UI	User Interface
URL	Uniform Resource Locator
Flask	Python Web Framework

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1 – Chapter

INTRODUCTION

1.1 About the Project

In today's digital world, financial management has become an important part of everyday life

People often face difficulties in managing expenses, maintaining financial records and monitoring spending habits. Traditional methods such as notebooks and spreadsheets are time-consuming and may lead to errors in expense calculation and management.

FinTrack: Family and Expense Management System is a web-based application developed to provide an efficient solution for managing both personal and family expenses. The system helps users record daily expenses, categorize spending and monitor budgets in an organized manner.

The application provides separate modules for personal expense management and family expense management. Users can add, edit and delete expenses, monitor monthly budgets and analyze spending patterns through charts and reports. The system also provides category-wise budget limits and smart alerts to help users avoid overspending.

One of the important features of FinTrack is family expense management, where multiple members can manage shared expenses together. The system supports expense splitting, which automatically divides expenses among selected family members, reducing manual calculations and confusion.

The project is developed using Python Flask, SQLite, HTML, CSS and JavaScript to provide a simple, efficient and user-friendly financial management system. The application also supports offline access, allowing users to manage expenses without internet connectivity.

1.2 Need for the System

Managing expenses manually is a difficult and time-consuming process. Many individuals and families find it hard to track daily expenses, manage budgets and maintain accurate financial records. Manual tracking methods such as notebooks or spreadsheets can lead to mistakes, data loss and poor financial planning.

In many cases, people are unable to identify where their money is being spent, which results in unnecessary expenses and poor budgeting habits. Families especially face difficulties in maintaining shared expenses,

dividing payments and tracking the spending of multiple members.

There is a need for a smart and user-friendly system that can simplify expense management and provide clear financial insights. FinTrack is developed to solve these problems by providing digital expense tracking, budget monitoring, spending analysis and professional reporting.

The system helps users improve financial planning and make better spending decisions through organized records, charts and alerts.

1.3 Goals of the Project

The main goal of FinTrack: Family and Expense Management System is to provide an efficient and organized platform for managing personal and family expenses.

The major goals of the project are:

- To provide an easy method for recording daily expenses.
- To manage both personal and family financial activities.
- To allow users to set monthly budgets and category-wise limits.
- To provide smart alerts when spending exceeds the budget.
- To support expense sharing among family members.
- To generate professional expense reports in PDF format.
- To analyze spending patterns using charts and graphical reports.
- To provide calendar-based expense tracking.
- To improve financial awareness and budgeting habits.

1.4 Problem Statement

Many people still use traditional methods such as notebooks, bills or spreadsheets for maintaining expenses. These methods are inefficient, difficult to maintain and may result in calculation errors or missing records.

Existing systems often focus only on personal expense tracking and lack support for family expense management, shared expense splitting, budget monitoring and expense analysis. Users also face difficulty in generating reports and understanding spending patterns.

Another major problem is the absence of category-wise budgeting and smart alerts, which makes it difficult for users to control overspending.

Therefore, there is a need for a smart and effective system that can manage expenses digitally, support family financial activities and provide better financial analysis. FinTrack is developed to address these problems through an organized and user-friendly platform.

1.5 Objectives of the System

The objectives of the system are:

- To develop a smart expense management system for individuals and families.
- To simplify expense recording and tracking.
- To provide budget management facilities.
- To support family expense sharing and split calculation.
- To generate professional reports for expense analysis.
- To improve spending management through graphical analysis.
- To provide secure login and profile management features.
- To offer offline functionality for easy access without internet.
-

1.6 Scope of the Project

The scope of FinTrack: Family and Expense Management System includes managing both personal and family expenses through a digital platform. The system allows users to maintain expense records, monitor spending habits and improve budgeting practices.

The project supports expense categorization, budget management, family expense sharing, calendar view and professional report generation. It also provides graphical analysis for better understanding of spending.

Chapter - 2

LITERATURE SURVEY

2.1 Introduction

Literature survey is an important phase in project development that involves studying existing systems, research papers, technologies and methods related to the project. It helps in understanding the strengths, weaknesses and working principles of already available systems and provides ideas for developing an improved solution.

In the field of expense management, many systems and applications have been developed to help users maintain financial records and monitor expenses. These systems provide features such as expense recording, budgeting and financial reporting. However, many existing systems have limitations in managing family expenses, shared financial activities and advanced expense analysis.

The study of previous research and existing applications helps in identifying the problems faced by users and understanding the need for better financial management systems. Through literature survey, it becomes easier to analyze how different systems handle expense tracking, budgeting and reporting.

For the development of FinTrack: Family and Expense Management System, various expense tracking applications, family financial management systems and budget monitoring systems were studied. This helped in understanding existing limitations and designing a better system with improved functionalities such as personal and family expense tracking, shared expense splitting, category-wise budgeting, expense analytics and professional report generation.

The literature survey also provides a foundation for selecting suitable technologies and methodologies required for project development. Based on the study, technologies such as Python Flask, SQLite, HTML, CSS and JavaScript were selected to develop an efficient, reliable and user-friendly expense management system.

The main purpose of this literature survey is to analyze existing solutions, understand their limitations and develop a more effective system that improves financial management for both individuals and families.

2.2 Study of Expense Management Systems

Expense management systems are software applications developed to help individuals and organizations track, organize and manage financial expenses efficiently. These systems reduce the complexity of manual expense recording and provide users with better control over their financial activities.

Traditional expense management methods mainly involve maintaining records using notebooks, receipts or spreadsheets. Although these methods are simple, they are often time-consuming, difficult to maintain and may result in errors or loss of important financial information. As financial activities increase, manual management becomes inefficient and difficult to analyze.

Modern expense management systems provide digital solutions for recording daily expenses, categorizing spending and maintaining financial records. These systems generally include features such as expense entry, category classification, monthly budgeting and report generation. Some applications also provide graphical analysis to help users understand spending behavior more effectively.

Several popular expense tracking applications are available today, allowing users to monitor expenses through mobile and web platforms. These systems help users maintain organized financial records and improve money management practices. However, many existing systems focus mainly on individual expense management and do not provide sufficient support for family expense tracking and shared financial activities.

Another important feature studied in expense management systems is expense categorization. Expenses are grouped into categories such as food, travel, bills, shopping and healthcare, allowing users to understand where money is spent most frequently. Budget tracking is also an essential function, where users can set spending limits and monitor expenses regularly.

During the study of expense management systems, it was observed that many applications have limitations such as lack of shared expense management, limited reporting functionality and insufficient financial analytics. These limitations highlighted the need for a more efficient system capable of managing both personal and family expenses in an integrated manner.

Based on this study, FinTrack: Family and Expense Management System was designed to overcome the limitations of traditional and existing expense management systems by providing advanced features such as family expense tracking, shared expense splitting, category-wise budgeting, spending analysis and professional report generation.

2.3 Family Financial Management Systems

Family financial management systems are designed to help multiple family members manage household expenses, shared budgets and overall financial activities in an organized manner. These systems provide a centralized platform where family members can monitor spending and maintain financial transparency.

In many families, expenses such as groceries, rent, electricity bills, school fees and transportation are shared among members. Managing these expenses manually often creates confusion regarding who paid the amount and how expenses should be divided. This may lead to poor financial planning and misunderstandings among family members.

Traditional financial management methods generally focus on individual spending and do not provide proper support for collective family expense tracking. Most expense tracking applications allow only single-user management, which makes it difficult to maintain a combined family expense record.

Modern family financial management systems aim to solve these problems by allowing multiple users to manage expenses under a common platform. Such systems provide features like expense sharing, member-wise expense tracking, budget management and spending analysis. Some systems also support expense splitting, where shared expenses are automatically divided among selected family members.

During the study of family financial management systems, it was observed that many existing solutions provide only basic expense tracking and lack advanced functionalities such as category-wise budgeting, graphical analysis and professional report generation. Many systems also fail to provide effective management for both personal and family expenses within a single platform.

To overcome these limitations, FinTrack: Family and Expense Management System includes a dedicated family expense management module where family members can collectively manage expenses. The system supports shared expense splitting, member-wise expense tracking, budget monitoring and spending analysis, making family financial management more efficient and organized.

Thus, the study of family financial management systems helped in understanding the importance of collaborative financial planning and influenced the development of features included in FinTrack.

2.4 Budget Monitoring Systems

Budget monitoring systems are developed to help users manage their income and expenses by setting financial limits and monitoring spending patterns. These systems play an important role in improving financial discipline and preventing unnecessary expenses.

A budget is a planned estimate of income and expenditure over a specific period of time. Without proper budgeting, individuals and families may face financial difficulties due to uncontrolled spending. Budget monitoring systems help users compare their actual spending with planned expenses and maintain financial balance.

Traditional methods of budget monitoring mainly include manual calculations using notebooks or spreadsheets. Although these methods are simple, they require continuous manual updates and may lead to calculation errors. Users often find it difficult to monitor spending regularly and identify when expenses exceed budget limits.

Modern budget monitoring systems provide digital solutions that allow users to set monthly or category-wise budgets and track expenses automatically. These systems notify users when spending reaches a warning level or exceeds the predefined limit. Such alerts help users make better financial decisions and avoid overspending.

During the study of budget monitoring systems, it was observed that many applications support only general budget tracking and lack detailed category-based monitoring. Some systems do not provide alerts or graphical analysis, making it difficult for users to understand spending habits clearly.

Another important feature of budget monitoring systems is expense categorization, where users can assign limits to categories such as food, travel, shopping and bills. This allows users to identify areas of high spending and control unnecessary expenses effectively.

Based on the study, FinTrack: Family and Expense Management System includes a dedicated budget management feature that supports monthly budget tracking, category-wise budget limits and smart alerts. The system automatically notifies users when spending approaches or exceeds the budget, helping them improve financial planning and spending control.

Thus, budget monitoring systems provided valuable insights for developing an efficient expense tracking system that promotes better money management and financial awareness.

2.5 Expense Analysis and Reporting Systems

Expense analysis and reporting systems are developed to help users understand their spending behavior through data analysis, charts and financial reports. These systems assist individuals and families in monitoring expenses, identifying spending patterns and making better financial decisions.

Traditional expense tracking methods mainly focus on recording expenses without providing detailed analysis or meaningful financial insights. Manual methods such as notebooks and spreadsheets make it difficult to analyze monthly spending, compare expenses and identify areas of unnecessary expenditure.

Modern expense analysis systems provide various analytical features such as graphs, charts and expense summaries. These visual representations help users understand where money is being spent most frequently and identify categories with higher expenses. Spending trends over weeks or months can also be analyzed for better financial planning.

Reporting systems are another important feature of expense management applications. Reports provide detailed summaries of expenses, budgets and spending activities in a structured format. Professional reports help users review financial records and maintain proper documentation of expenses.

Many modern systems generate reports in formats such as PDF and include details such as category-wise expenses, monthly spending, payment methods and expense summaries. Some applications also include visual reports with graphs for better understanding.

During the study of expense analysis and reporting systems, it was observed that many existing systems provide only basic reporting features and lack advanced analytics, budget comparison and detailed financial suggestions. Several systems also fail to provide professional report formats suitable for maintaining financial records.

To overcome these limitations, FinTrack: Family and Expense Management System includes graphical expense analysis using charts and monthly comparisons. The system also generates professional PDF reports containing expense details, tables and spending summaries. These features help users monitor financial activities more effectively and improve budgeting habits.

Thus, the study of expense analysis and reporting systems helped in designing a better system that provides meaningful financial insights and organized reporting for both personal and family expense management.

2.6 Comparative Analysis of Existing Systems

A comparative analysis of existing systems was carried out to understand the features, advantages and limitations of currently available expense management methods and applications. This study helped in identifying the gaps in traditional systems and the need for a more efficient and user-friendly expense management solution.

Traditional methods such as notebooks and spreadsheets are commonly used for maintaining expense records. These methods are simple but require manual effort for calculation, maintenance and analysis. They are time-consuming and may lead to calculation errors, missing records and poor financial management.

Several modern expense tracking applications provide features such as expense recording, budgeting and report generation. Although these systems are useful for personal expense tracking, many of them have limitations in supporting family financial management, shared expense splitting and category-wise budgeting.

Some existing systems provide graphical analysis and financial reports, but they often lack customization, advanced budgeting features and proper support for family-based financial activities. Many systems also require internet connectivity and subscription-based services, limiting accessibility for users.

The comparative study showed that there is a need for a system that combines personal expense tracking, family expense management, budget monitoring and professional reporting within a single platform.

The following table presents a comparison between existing systems and the proposed system (FinTrack)

Features	Existing Systems	FinTrack
Personal Expense Tracking	Available	Available
Family Expense Management	Limited / Not Available	Available
Shared Expense Splitting	Not Available	Available
Category-wise Budgeting	Limited	Available
Smart Budget Alerts	Limited	Available
Graphical Analysis	Basic	Advanced
Professional PDF Reports	Limited	Available
Offline Access	Limited	Available

From the comparison, it is clear that existing systems do not fully support combined personal and family financial management. Therefore, FinTrack: Family and Expense Management **System** is proposed as an improved solution that provides advanced features for efficient expense tracking, budgeting and financial analysis.

2.7 Summary

The literature survey provided a detailed understanding of various expense management systems, family financial management systems, budget monitoring systems and expense analysis methods. Different existing approaches and technologies were studied to understand their functionalities, advantages and limitations.

The study revealed that traditional expense management methods such as notebooks and spreadsheets are time-consuming, difficult to maintain and prone to errors. Modern expense tracking systems provide digital solutions for recording expenses and budget monitoring, but many applications mainly focus on individual expense management and provide limited support for family financial activities.

It was also observed that many existing systems lack important features such as shared expense splitting, category-wise budgeting, professional report generation and detailed financial analytics. Several applications provide only basic expense tracking and do not support effective family expense management.

The literature survey helped in understanding the need for an integrated system that can manage both personal and family expenses efficiently. It also provided valuable insights for selecting suitable technologies and designing better functionalities for the project.

Based on the findings of the literature survey, FinTrack: Family and Expense Management System was developed to overcome the limitations of existing systems. The proposed system provides advanced features such as personal and family expense management, budget monitoring, smart alerts, expense analysis, shared expense splitting and professional report generation.

Thus, the literature survey played an important role in understanding existing solutions and forming the **3** –

Chapter - 3

EXISTING SYSTEM

3. 1 Existing Expense Management System

The existing expense management systems mainly depend on traditional methods such as notebooks, receipts, manual calculations and spreadsheet applications for maintaining financial records. Many individuals and families still use these methods to track daily expenses and manage budgets. Although these methods are simple and commonly used, they are often inefficient and difficult to maintain over long periods.

Manual expense tracking requires users to record every expense individually, which is time-consuming and may result in errors or missing records. Maintaining accurate calculations using notebooks or spreadsheets becomes difficult when expenses increase. In many situations, users fail to monitor their spending patterns properly, which leads to poor budgeting and unnecessary expenses.

Several modern expense tracking applications are available that provide digital expense management features such as recording expenses, categorizing spending and generating reports. These systems help users maintain financial records more efficiently compared to traditional methods. However, many of these applications mainly focus on personal expense management and provide only limited support for family expense tracking.

Most existing systems lack proper support for family financial management, where multiple family members can collectively manage shared expenses. Users often face difficulties in maintaining common family expenses and calculating how much each member has contributed or spent. Shared expense management is generally performed manually, which creates confusion and increases calculation complexity.

Another major limitation of existing systems is the lack of shared expense splitting functionality. When multiple people share an expense, existing systems do not automatically

divide the amount among selected members. Users must manually calculate and distribute expenses, which is time-consuming and may lead to incorrect calculations.

Many expense tracking systems also provide limited support for budget management. While some applications allow users to set general monthly budgets, they often do not include category-wise budget limits for expenses such as food, travel, shopping and bills. As a result, users find it difficult to control overspending in specific categories.

Existing systems also provide only basic expense analysis and reporting features. In many cases, users do not receive proper graphical analysis, monthly spending comparisons or professional reports. This makes it difficult to understand spending behavior and identify areas where money can be saved.

Another limitation is that many modern systems require internet connectivity and depend on online services or subscription-based platforms. This reduces accessibility for users who require a simple offline financial management solution.

3.2 Limitations of Existing System

The major limitations of existing systems are listed below:

- No proper family expense tracking system.
- No shared expense splitting among family members.
- Limited budget monitoring features.
- Lack of category-wise budgeting and smart alerts.
- Basic or limited expense analysis and reporting.
- Dependency on internet connectivity in some systems.

Due to these limitations, there is a need for a better and more efficient expense management system that supports both personal and family financial management with improved

budgeting, reporting and analytical capabilities. These drawbacks led to the development of FinTrack: Family and Expense Management System, which provides an integrated and user-friendly solution for managing expenses effectively.

Chapter - 4

PROPOSED SYSTEM

4.1 Overview of Proposed System

FinTrack: Family and Expense Management System is proposed as an advanced and efficient solution for managing both personal and family expenses. The system is designed to overcome the limitations found in existing expense management methods by providing a smart, organized and user-friendly platform for financial management.

The proposed system allows users to digitally record and manage expenses in a systematic manner. Unlike traditional expense management methods, FinTrack provides a centralized platform where users can monitor spending activities, set budgets and analyze financial records efficiently. The system reduces manual effort and helps users maintain accurate expense records.

One of the important features of the proposed system is personal expense management, where users can add, update and delete expenses according to different categories such as food, travel, shopping, healthcare, bills and entertainment. This helps users organize their financial activities and understand spending behavior more effectively.

Another major feature of the system is family expense management, which allows multiple family members to manage expenses together. Family expenses can be recorded and monitored collectively, making it easier to maintain financial transparency within the family.

The proposed system also provides shared expense splitting functionality, where expenses can be divided automatically among selected family members. This reduces manual calculations and avoids confusion regarding expense sharing.

To improve financial planning, the system includes budget management features, allowing users to set monthly spending limits. Users can also define category-wise budget limits to control spending in areas such as food, shopping and travel. The system provides smart alerts and notifications when expenses approach or exceed the budget limit.

FinTrack also provides expense analysis and visualization features through graphs and charts. Users can analyze monthly spending patterns, category-wise expenses and budget utilization. This graphical representation helps users better understand their financial behavior and make informed spending decisions.

Another important feature of the proposed system is professional report generation. The system generates expense reports in PDF format, containing expense details, tables and financial summaries. This helps users maintain financial documentation in an organized way.

The system also supports calendar-based expense tracking, where users can view expenses according to specific dates for easier monitoring and analysis.

Unlike many existing systems, FinTrack works offline, allowing users to access the system without internet connectivity. The project is developed using a local Flask server and SQLite database, making it lightweight, reliable and easy to use.

4.2 Advantages of Proposed System

The following are the major advantages of the proposed system:

- Supports both personal and family expense management.
- Provides automatic shared expense splitting.
- Allows category-wise budget management.
- Generates smart alerts for overspending.
- Provides charts and graphical spending analysis.
- Generates professional PDF reports.
- Supports offline functionality.
- Offers secure login and profile management.

Chapter - 5

SYSTEM REQUIREMENTS

5.1 Introduction

System requirements are the necessary hardware and software specifications required for the successful development and execution of a project. These requirements ensure that the system functions efficiently and provides smooth performance without technical issues.

For the development of FinTrack: Family and Expense Management System, appropriate hardware and software resources are required to support application development, database management and system execution. The system is developed as a web-based application using modern technologies that provide reliability, efficiency and user-friendly interaction.

The hardware requirements define the physical components needed to run the application, such as processor, memory and storage capacity. Similarly, software requirements include programming languages, frameworks, database systems and development tools used for implementation.

Apart from hardware and software specifications, the project also includes functional requirements, which define the operations the system must perform, and non-functional requirements, which describe performance, security and usability aspects of the application.

A feasibility study is also conducted to analyze whether the project is practical and beneficial in terms of technical implementation, operational use and cost effectiveness. Thus, system requirements play an important role in ensuring the successful development and proper functioning of FinTrack: Family and Expense Management System.

5.2 Hardware Requirements

Hardware requirements refer to the physical components required for the proper functioning and execution of the system. The following hardware specifications are sufficient for the smooth operation of FinTrack: Family and Expense Management System.

Hardware Component Specification

Processor	Intel Core i3 or Above
RAM	4 GB or Above
Hard Disk	20 GB Free Space
Monitor	15-inch or Above
Keyboard	Standard Keyboard
Mouse	Standard Mouse
System Type	64-bit Operating System

The above hardware configuration is adequate for running the application efficiently without performance issues. Since the project is developed as a lightweight web-based system using Flask and SQLite, it does not require high-end hardware resources.

The system can operate smoothly on standard desktop or laptop computers commonly used in educational institutions and personal environments.

5.3 Software Requirements

Software requirements refer to the programming languages, frameworks, development tools and operating systems required for the successful development and execution of the project. The following software components are used in the development of FinTrack: Family and Expense Management System.

Software Component Specification

Operating System	Windows 10 / Windows 11
Programming Language	Python
Framework	Flask
Frontend Technologies	HTML, CSS, JavaScript

Software Component	Specification
Database	SQLite
IDE	Visual Studio Code (VS Code)
Browser	Google Chrome / Microsoft Edge
Libraries Used	Flask, SQLAlchemy, ReportLab, Matplotlib
Version Control	Local File Management

The project is developed using Python because of its simple syntax, flexibility and extensive library support. Flask is used as the web framework for backend development due to its lightweight and flexible nature.

The frontend interface is designed using HTML, CSS and JavaScript, providing a user-friendly and interactive environment. SQLite is used as the database system for storing user information, expenses and budget details efficiently.

Visual Studio Code (VS Code) is used as the development environment for coding, debugging and testing the application. The system is executed through modern web browsers such as Google Chrome or Microsoft Edge for better compatibility and performance.

The application also uses libraries such as Report Lab for PDF report generation, Matplotlib for graphical visualization and SQL Alchemy for database operations.

These software requirements ensure efficient development, smooth execution and reliable performance of the proposed system.

5.4 Functional Requirements

Functional requirements describe the operations and services that the system must perform to meet user needs. These requirements define the major functionalities of FinTrack: Family and Expense Management System and explain how the system interacts with users.

The functional requirements of the proposed system are as follows:

1.User Authentication

The system should allow users to register and log in securely using valid credentials. Users should also be able to change passwords and manage their profiles.

2.Personal Expense Management

The system should allow users to add, edit, update and delete personal expenses. Users should be able to categorize expenses and track spending activities.

3.Family Expense Management

The system should support family expense tracking where multiple family members can record and manage shared expenses collectively.

4. Shared Expense Splitting

The system should automatically divide shared expenses among selected family members to simplify expense calculation.

5. Budget Management

The system should allow users to set monthly budgets and monitor spending according to budget limits.

6. Category-wise Budget Management

The system should support category-based spending limits for categories such as food, travel, shopping and bills.

7. Expense Analysis

The system should provide graphical analysis through charts and monthly spending comparisons to help users understand spending patterns.

8. Report Generation

The system should generate professional expense reports in PDF format containing expense details and summaries.

9. Calendar View

The system should provide a calendar-based view for tracking expenses according to dates.

10. Offline Access

The system should function without internet connectivity using a local database and local server.

These functional requirements ensure that the system performs all necessary operations efficiently and

provides a complete expense management solution for both individuals and families.

5.5 Non-Functional Requirements

Non-functional requirements define the quality, performance and operational characteristics of the system. These requirements ensure that FinTrack: Family and Expense Management System performs efficiently, securely and reliably while providing a better user experience.

The non-functional requirements of the system are as follows:

1. Performance

The system should provide fast response time while adding, updating and retrieving expense records. The application should operate smoothly without delays.

2. Reliability

The system should function consistently without failures and maintain accurate financial records for users.

3. Security

The system should provide secure user authentication through login credentials and password management to prevent unauthorized access.

4. Usability

The system should provide a simple, user-friendly and interactive interface so that users can easily understand and use the application without technical knowledge.

5. Scalability

The system should be flexible enough to allow future improvements and additional features such as cloud storage, mobile application support and AI-based expense prediction.

6. Availability

The application should be available whenever required and should support offline functionality without internet dependency.

7. Maintainability

The system should be easy to update, modify and maintain for future enhancements and bug fixing.

8. Compatibility

The system should run efficiently on standard operating systems and modern web browsers such as Google

Chrome and Microsoft Edge.

9. Data Integrity

The system should store accurate and consistent financial data without loss or duplication of expense records.

These non-functional requirements help ensure better performance, security, reliability and user satisfaction in the proposed system.

5.6 Feasibility Study

A feasibility study is conducted to determine whether the proposed project is practical, beneficial and technically possible to implement. It helps in analyzing the viability of the system in terms of technology, operation and cost. The feasibility study of FinTrack: Family and Expense Management System is explained below.

1. Technical Feasibility

Technical feasibility refers to the availability of the required technology, tools and resources needed to develop the system. The proposed system is developed using Python Flask, SQLite, HTML, CSS and JavaScript, which are lightweight and easily available technologies. The required software and hardware are easily accessible, making the project technically feasible.

2. Economic Feasibility

Economic feasibility determines whether the project can be developed at a reasonable cost. The proposed system uses open-source technologies such as Python, Flask and SQLite, which do not require expensive licenses or software costs. Therefore, the project can be developed and maintained economically with minimum expenses.

3. Operational Feasibility

Operational feasibility refers to how effectively the system can be used by end users. FinTrack provides a simple and user-friendly interface, making it easy for users to manage personal and family expenses without technical difficulties. The system is designed to improve financial management and provide better spending control.

4. Schedule Feasibility

Schedule feasibility checks whether the project can be completed within the available time period. The project was developed module by module, including authentication, expense management, budgeting, reporting and analysis features. Since proper planning and development strategies were followed, the project was completed within the scheduled time.

Thus, the feasibility study confirms that FinTrack: Family and Expense Management System is technically possible, economically affordable, operationally effective and feasible for implementation.

Chapter - 6

SYSTEM DESIGN

6.1 Introduction

System design is an important phase in project development that defines the structure, components and overall working of the proposed system. It provides a detailed representation of how the system functions, how different modules interact and how data is processed within the application.

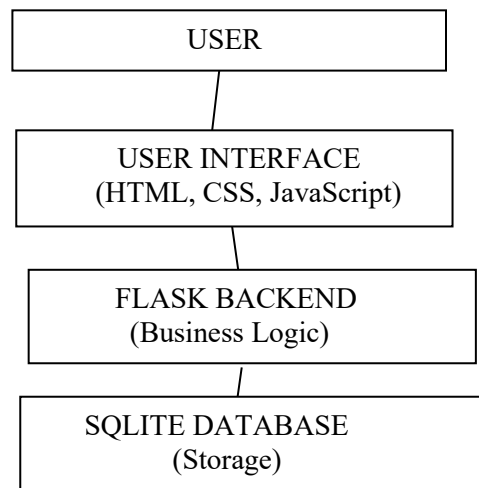
The system design of FinTrack: Family and Expense Management System focuses on creating an organized, user-friendly and efficient platform for managing personal and family expenses. The design phase helps in understanding the architecture, workflow and internal structure of the system before implementation.

The proposed system is designed using a modular approach, where different functionalities such as authentication, expense management, budgeting, report generation and expense analysis are separated into individual modules for better efficiency and maintainability. Each module performs specific tasks and works together to provide smooth functioning of the system.

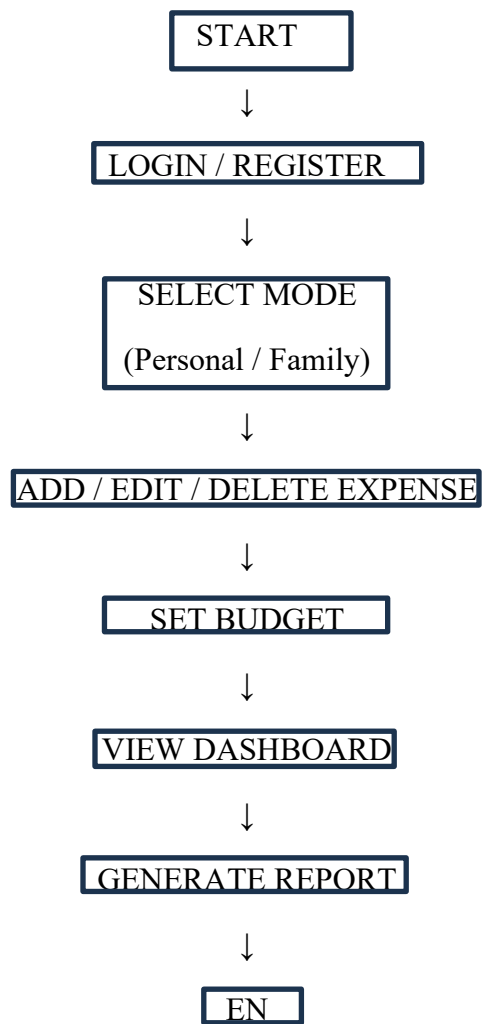
System design also includes the creation of various diagrams such as system architecture, flowchart, use case diagram and entity relationship (ER) diagram. These diagrams help in understanding the interaction between users and the system, the movement of data and the relationship between database components.

The database design plays an important role in storing and managing user details, expenses, budgets and reports effectively. SQLite database is used for maintaining accurate and structured data records.

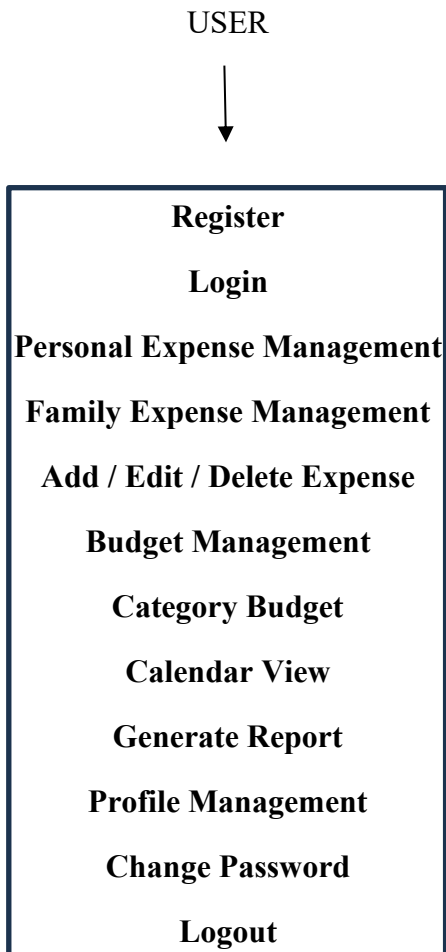
6.2 System Architecture



6.3 Flow Chart



6.4 Use Case Diagram



6.5 Database Design

Database design is an important part of system development that helps in storing, organizing and managing information efficiently. A database is used to maintain records in a structured format so that data can be easily stored, retrieved, updated and deleted whenever required.

In FinTrack: Family and Expense Management System, the database is designed using SQLite, which is a lightweight and efficient database management system. SQLite is chosen because it is simple, reliable and suitable for offline applications.

The database stores information related to users, expenses, family members, budgets and reports. Different tables are created to manage various functionalities of the system effectively. The relationship between

tables helps maintain proper organization and data consistency.

The major tables used in the system are explained below

1. User Table

The User table stores information related to registered users.

Field Name	Data Type	Description
id	Integer	Unique user ID
name	String	Name of user
email	String	Email ID
password	String	User password
family_id	Integer	Family group ID

2. Personal Expense Table

The Personal Expense table stores personal expense details of users.

Field Name	Data Type	Description
id	Integer	Expense ID
user_id	Integer	User reference ID
category	String	Expense category
amount	Float	Expense amount
date	String	Expense date
payment_mode	String	Payment method
notes	String	Additional notes

3. Family Expense Table

The Family Expense table stores family expense details and shared expenses.

Field Name	Data Type	Description
id	Integer	Expense ID
user_id	Integer	User reference ID
member	String	Family member name
category	String	Expense category
amount	Float	Expense amount
is_shared	Boolean	Shared expense status
paid_by	String	Paid member
split_members	String	Shared members
date	String	Expense date
payment_mode	String	Payment method
notes	String	Additional notes

4. Budget Table

The Budget table stores monthly budget information.

Field Name	Data Type	Description
id	Integer	Budget ID
user_id	Integer	User ID
month	String	Budget month
amount	Float	Budget amount
type	String	Personal/Family

5. Category Budget Table

The Category Budget table stores category-wise budget limits

Field Name	Data Type	Description
id	Integer	Category ID
user_id	Integer	User ID
category	String	Expense category
amount	Float	Budget amount
mode	String	Personal/Family

6. Family Member Table

The Family Member table stores family member information.

Field Name	Data Type	Description
id	Integer	Member ID
family_id	Integer	Family ID
name	String	Member name
relation	String	Relationship

7. Member Budget Table

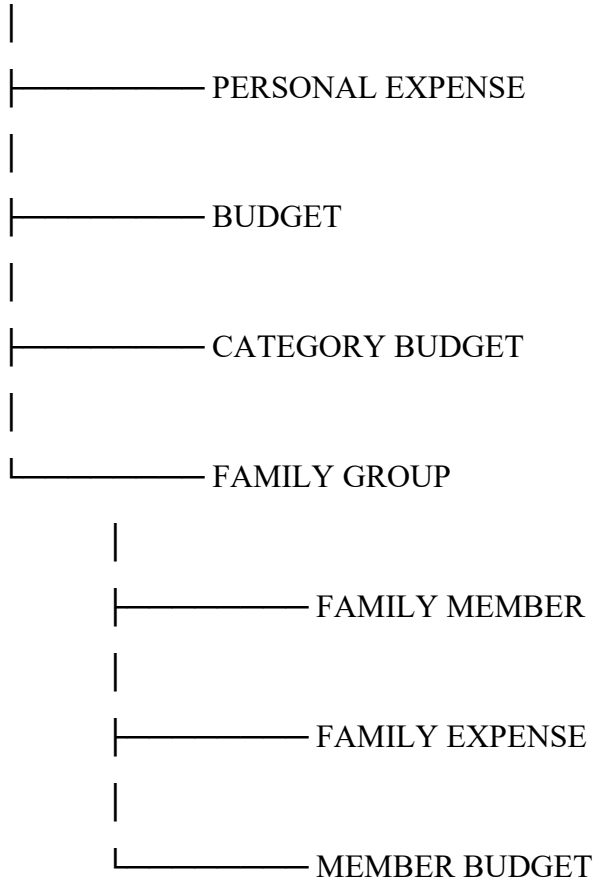
The Member Budget table stores individual family member budget details.

Field Name	Data Type	Description
id	Integer	Budget ID
member_name	String	Member name
family_id	Integer	Family ID
month	String	Month
amount	Float	Budget amount

The database design ensures proper storage and management of financial records. It helps maintain data accuracy, consistency and efficient retrieval of information, thereby improving the overall performance of

6.6 Entity Relationship (ER) Diagram

USER



Explanation of Relationships

- One User can have many Personal Expenses.
- One User can create many Budgets.
- One User can define many Category Budgets.
- One Family Group can contain multiple Family Members.
- One Family Group can have multiple Family Expenses.
- One Family Group can maintain multiple Member Budgets.

6.7 Module Description

The FinTrack: Family and Expense Management System is divided into different modules to ensure proper organization, efficient functioning and easy maintenance of the application. Each module performs specific operations and contributes to the overall working of the system.

The major modules of the system are explained below:

1. User Authentication Module

The User Authentication Module is responsible for managing user registration, login and profile security. It verifies user credentials and allows only authorized users to access the system. This module also provides password change functionality and profile management features.

2. Personal Expense Management Module

The Personal Expense Management Module helps users manage their personal financial activities. Users can add, edit, update and delete expenses according to different categories such as food, travel, shopping and bills. The module also displays spending details through dashboards and charts.

3. Family Expense Management Module

The Family Expense Management Module allows multiple family members to manage expenses collectively. Users can create or join family groups and maintain common financial records. This module supports family-based expense tracking for better financial transparency.

4. Shared Expense Splitting Module

This module allows users to divide shared expenses among selected family members automatically. When an expense is shared, the system calculates and distributes the expense amount equally among members, reducing manual calculations.

5. Budget Management Module

The Budget Management Module helps users set monthly budgets and monitor spending activities. The system compares expenses with budget limits and helps users control spending effectively.

6. Category Budget Module

The Category Budget Module allows users to define spending limits for different categories such as food, shopping, travel and healthcare. The system provides warnings and alerts when category spending approaches or exceeds the limit.

7. Expense Analysis Module

The Expense Analysis Module provides graphical representation of expenses through charts and monthly comparisons. Users can analyze spending patterns and identify categories with higher expenses.

8. Report Generation Module

The Report Generation Module generates professional PDF reports containing expense details, tables and spending summaries. This module helps users maintain financial records in an organized manner.

9. Calendar View Module

The Calendar View Module allows users to view expenses based on dates. It helps users track financial activities on specific days and improves expense monitoring.

10. Profile Management Module

The Profile Management Module allows users to manage personal details and change passwords securely. It helps maintain account security and user information.

Thus, the modular design of the system improves efficiency, maintainability and usability of FinTrack: Family and Expense Management System

Chapter - 7

SYSTEM IMPLEMENTATION

7.1 Introduction

System implementation is an important phase in project development where the designed system is converted into a working application. It involves integrating various modules, database operations and user interface components to ensure proper functioning of the system. The implementation phase helps in transforming theoretical design into a practical and usable solution.

The implementation of FinTrack: Family and Expense Management System is carried out using Python Flask, SQLite, HTML, CSS and JavaScript. The system is developed using a modular approach, where different functionalities are implemented separately and later integrated to provide smooth operation of the application.

The backend of the system is developed using the Flask framework, which handles application logic, user authentication, expense management and report generation. The frontend interface is designed using HTML, CSS and JavaScript, providing an interactive and user-friendly environment for users.

The system uses SQLite database to store user details, expense records, budgets and family information efficiently. Database operations such as insertion, deletion, updating and retrieval of records are performed using SQLAlchemy for better database management.

The implementation of the system includes several important modules such as user authentication, personal expense management, family expense tracking, budget management, expense analysis, report generation and profile management. Each module is developed and tested separately to ensure proper functionality.

Additionally, the project supports offline execution, where the application can run without internet connectivity using a local Flask server and local SQLite database. The project is also converted into an executable application using PyInstaller, making it easy to run directly on systems without requiring manual setup.

Thus, the implementation phase ensures that FinTrack: Family and Expense Management System functions effectively and provides an organized solution for managing personal and family financial activities.

7.2 User Authentication Module

The User Authentication Module is responsible for secure user access to the system. It allows users to register new accounts and log in using valid credentials. The system verifies login details and redirects users to the dashboard after successful authentication. Users can also change passwords and manage profiles securely.

Login

[New user? Register](#)

Register

[Already have account? Login](#)

7.3 Personal Expense Management Module

This module allows users to manage personal expenses efficiently. Users can add, edit, update and delete expenses. Expenses are categorized and displayed in the personal dashboard for better financial tracking

Add Expense

 Food

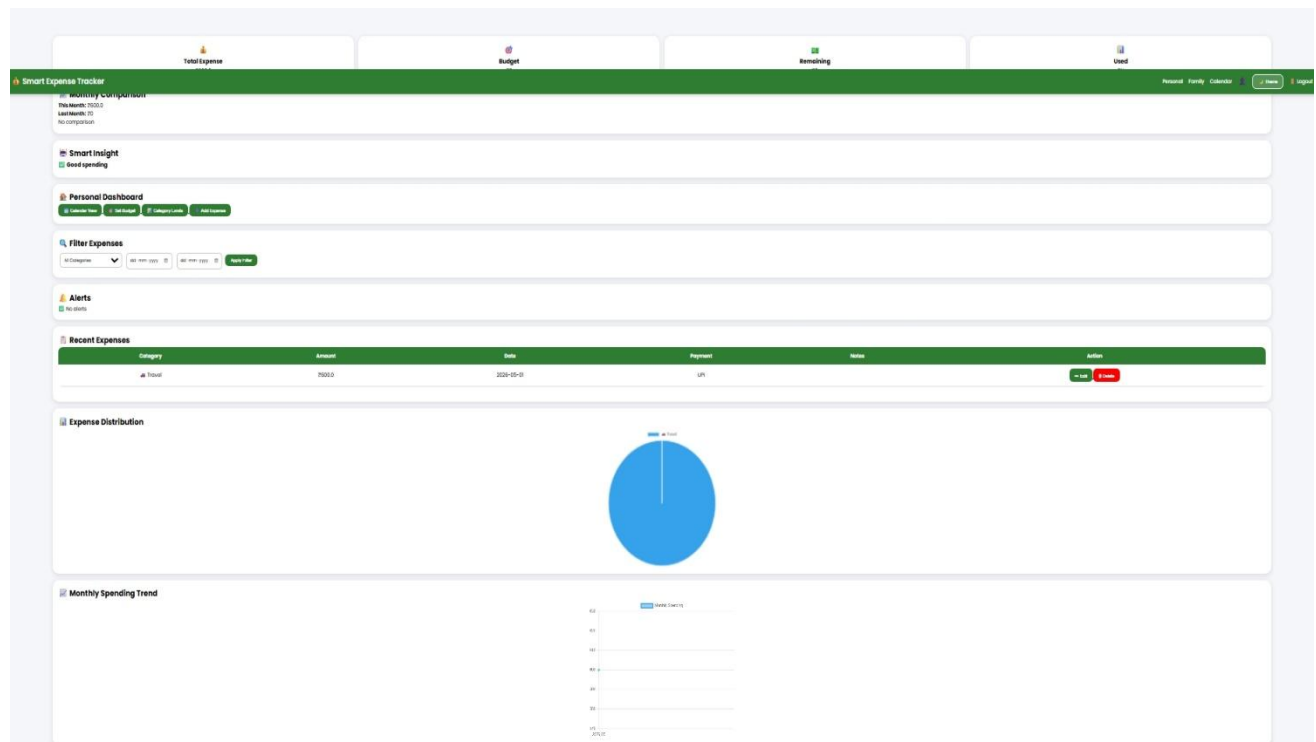
500

11-05-2026

Debit Card

Lunch

Add



7.4 Family Expense Management Module

The Family Expense Management Module helps family members manage expenses collectively. Users can create or join family groups and record shared expenses. The system also supports automatic expense splitting.

Add Family Expense

charan (son) ▼

charan (son) ▼

Travel ▼

1000

Split Expense

Paid By: charan ▼

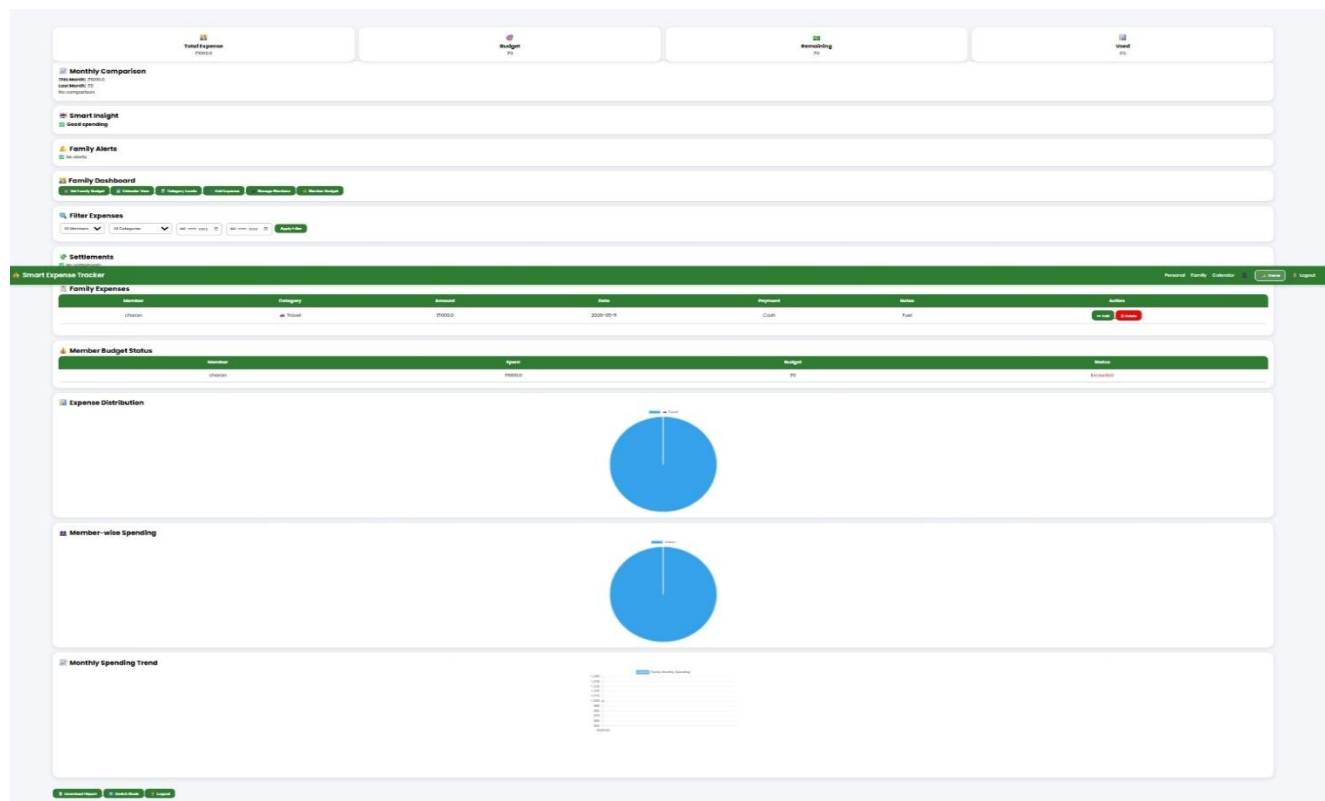
Split Between:
☐ charan

11-05-2026

UPI ▼

Fuel

Add Expense



7.5 Budget Management Module

This module allows users to set monthly budgets and category-wise spending limits. The system provides alerts when expenses exceed budget limits.

Set Budget

Save Budget

Category Budget

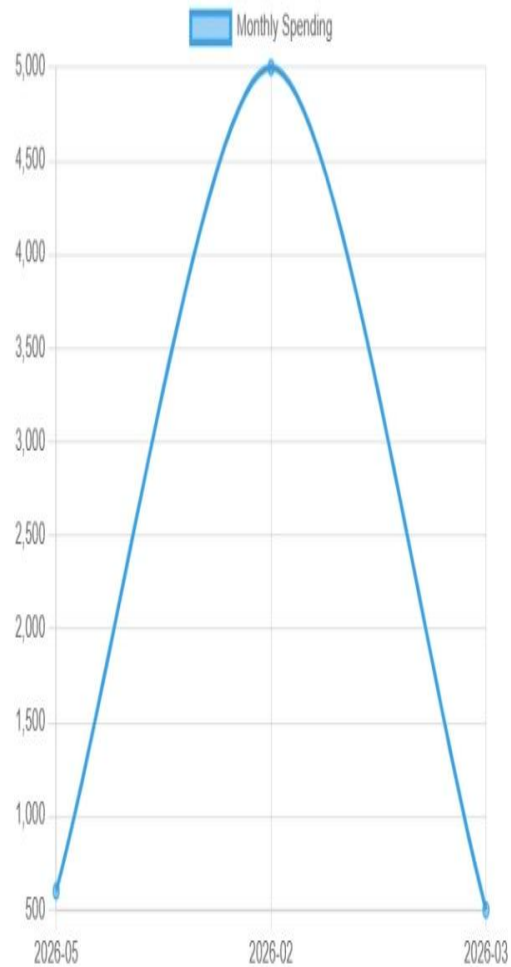
 Medical 

Set Limit

7.6 Expense Analysis and Dashboard Module

This module provides expense analysis using charts, monthly comparisons and dashboard summaries. It helps users understand spending behavior and monitor financial activities.

Monthly Spending Trend



7.7 Report Generation Module

This module generates professional PDF reports containing expense details, spending summaries and financial analysis. Reports help users maintain organized financial records.

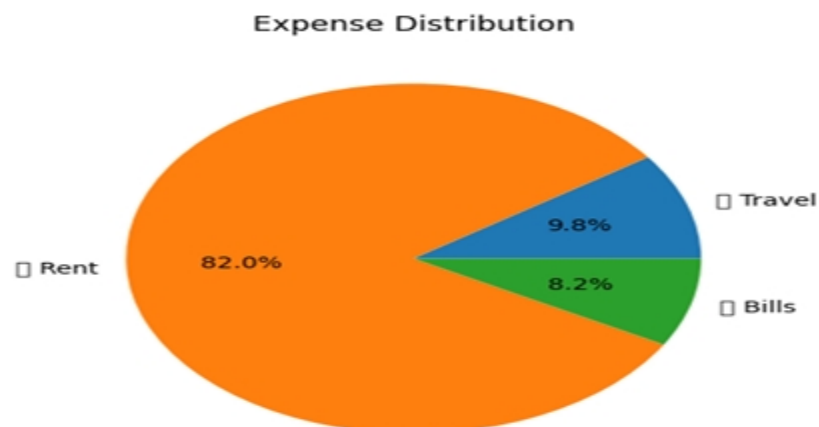
Smart Expense Tracker Report

Personal Expense Report

Total Expense: ■6100.0

Category	Amount	Date	Payment
■ Travel	600.0	2026-05-01	UPI
■ Rent	5000.0	2026-02-05	Debit Card
■ Bills	500.0	2026-03-12	Cash

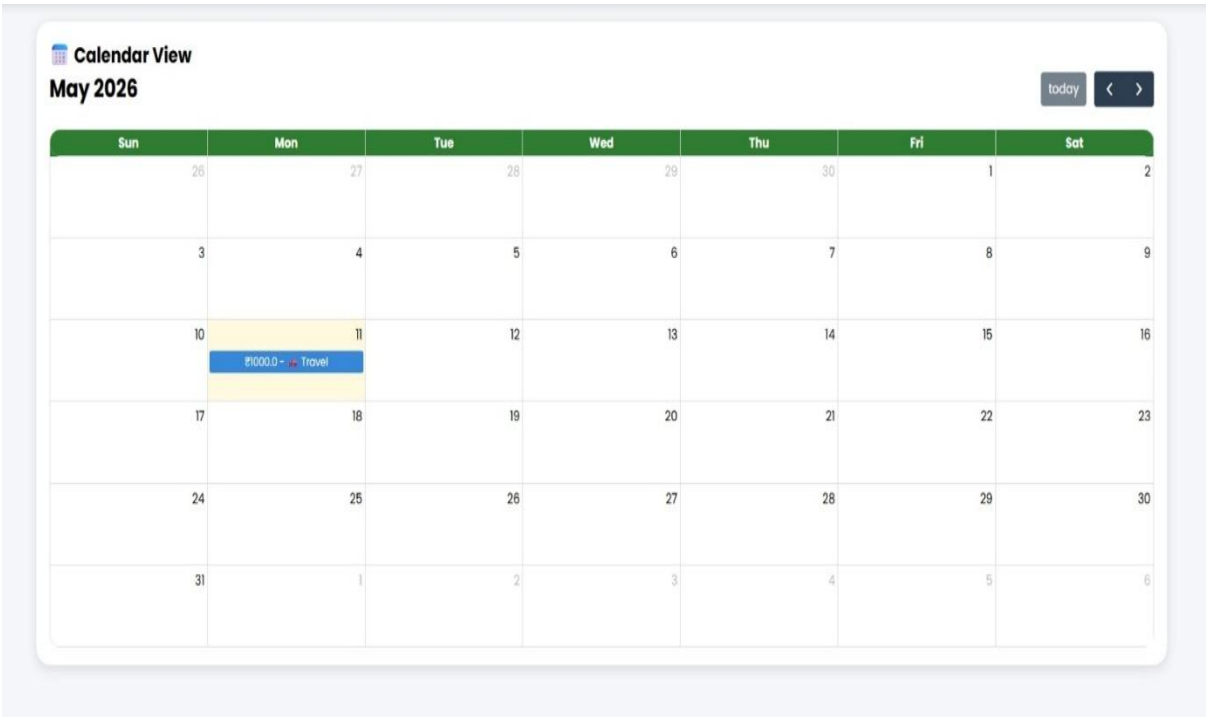
Expense Distribution



Suggestion: Highest spending is on ■ Rent. Try reducing it.

7.8 Calendar View and Profile Module

This module allows users to track expenses using a calendar view and manage profile information securely.



Chapter - 8

TESTING

8.1 Introduction

Testing is an important phase in software development that ensures the proper functioning, accuracy and reliability of the system. It helps identify errors, verify system performance and ensure that all functionalities work according to requirements.

The testing phase of FinTrack: Family and Expense Management System was carried out to verify the correctness of different modules such as login authentication, expense management, budgeting, report generation and family expense tracking. Each module was tested individually and collectively to ensure smooth system performance.

The purpose of testing is to ensure that the application performs efficiently without errors and provides correct results based on user input. Various testing methods such as Unit Testing, Integration Testing and System Testing were performed to validate the functionality of the proposed system.

Through proper testing, the system was verified to ensure reliability, usability and accurate financial record management.

8.2 Unit Testing

Unit testing is the process of testing individual modules or components of a system separately to verify whether they function correctly. In FinTrack: Family and Expense Management System, each module was

Login Module	Enter valid login credentials	User login successful	Pass
Registration Module	Register new user	Account created successfully	Pass
Personal Expense Module	Add personal expense	Expense added successfully	Pass
Family Expense Module	Add family expense	Expense recorded	Pass
Budget Module	Set monthly budget	Budget saved	Pass
Category Budget Module	Set category limit	Category budget updated	Pass
Report Module	Generate PDF report	Report downloaded successfully	Pass
Profile Module	Change password	Password updated	Pass

tested independently to ensure proper execution and error-free performance.

Different functionalities such as login authentication, expense management, budgeting and report generation were tested individually to confirm that the expected outputs were produced successfully.

The following table shows the unit testing performed in the system:

8.3 Integration Testing

Integration testing is the process of testing multiple modules together to ensure that they work properly as a combined system. It verifies whether data is transferred correctly between modules and whether the overall application performs efficiently without errors.

In FinTrack: Family and Expense Management System, integration testing was performed to ensure proper communication between modules such as authentication, expense management, budgeting, report generation and profile management.

The system was tested to verify that after successful login, users could access dashboards, add expenses, set budgets, generate reports and manage profiles without any issues. Database interaction between different modules was also tested to ensure proper data storage and

The following table shows the integration testing carried out in the system:

Integrated Modules	Test Description	Expected Result	Status
Login + Dashboard	Login and access dashboard	Dashboard displayed successfully	Pass
Expense + Database	Add expense and store data	Expense saved in database	Pass
Budget + Alerts	Set budget and track limit	Alert displayed correctly	Pass
Dashboard + Charts	Expense update in dashboard	Graph updated successfully	Pass
Report + Expense Data	Generate report using stored expenses	PDF generated successfully	Pass
Profile + Authentication	Change password and login again	Updated login successful	Pass

The integration testing results confirmed that all modules were properly connected and worked efficiently together without data inconsistency or system failure.

8.4 System Testing

System testing is the process of testing the complete application as a whole to ensure that all functionalities work correctly according to the project requirements. It helps verify the performance, reliability and usability of the system in a real working environment.

In FinTrack: Family and Expense Management System, system testing was conducted after integrating all modules successfully. The complete application was tested to ensure proper execution of functionalities such as user authentication, personal expense tracking, family expense management, budgeting, report generation and profile management.

The system was tested with different user inputs to verify whether accurate outputs were generated. It was also tested for data storage, expense calculations, budget alerts and graphical analysis to ensure proper system performance.

The following table shows the system testing carried out in the application:

Test Scenario	Input Given	Expected Result	Status
User Login	Valid credentials	Successful login	Pass
Add Expense	Expense details	Expense added	Pass
Edit Expense	Update amount/category	Expense updated	Pass
Delete Expense	Delete record	Expense removed	Pass
Set Budget	Monthly amount	Budget stored	Pass
Generate Report	Download report	PDF generated	Pass
Calendar View	Select date	Expense displayed	Pass
Change Password	Old and new password	Password updated	Pass

The results of system testing confirmed that the complete application functions efficiently and meets the intended objectives of the project. No major errors were found during testing, and the system performed reliably under normal usage conditions.

8.5 Test Cases and Results

Test cases are used to verify whether the system performs correctly for different user inputs and scenarios. The testing process ensures that the system produces expected results and functions properly without errors.

In FinTrack: Family and Expense Management System, different test cases were executed to validate major functionalities such as login, expense management, budgeting, report generation and profile management.

The following table shows the test cases and their results:

Test ID	Case	Test Scenario	Input	Expected Result	Actual Result	Status
TC01		User Login	Valid credentials	Login successful	Login successful	Pass
TC02		User Registration	User details	Account created	Account created	Pass
TC03		Add Personal Expense	Expense details	Expense added	Expense added	Pass
TC04		Add Family Expense	Family expense details	Expense recorded	Expense recorded	Pass
TC05		Set Monthly Budget	Budget amount	Budget saved	Budget saved	Pass
TC06		Set Category Budget	Category limit	Budget updated	Budget updated	Pass
TC07		Generate PDF Report	Click report option	Report generated	Report generated	Pass
TC08		View Dashboard	Expense records	Dashboard displayed	Dashboard displayed	Pass
TC09		Change Password	Password details	Password changed	Password changed	Pass
TC10		Calendar View	Select date	Expense displayed	Expense displayed	Pass

The test case results show that all important functionalities of the system performed successfully and produced the expected outcomes. The testing process confirmed that the system is reliable, accurate and

Chapter - 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The FinTrack: Family and Expense Management System was successfully designed and developed to provide an efficient solution for managing personal and family expenses. The system helps users record daily expenses, monitor spending patterns and maintain financial discipline in an organized manner.

The project overcomes many limitations of traditional expense management methods by providing digital expense tracking, budget monitoring and financial analysis. Users can manage both personal and family expenses through a single platform, making financial management easier and more efficient.

The system provides several useful features such as expense categorization, monthly budget management, category-wise spending limits, shared expense splitting, calendar view and professional PDF report generation. Graphical analysis and dashboard summaries also help users understand spending behavior and improve financial planning.

The application was developed using Python Flask, SQLite, HTML, CSS and JavaScript, providing a lightweight, reliable and user-friendly system. The project also supports offline execution, making it accessible without internet connectivity.

Testing results confirmed that all modules function correctly and provide accurate outputs without major errors. Thus, FinTrack: Family and Expense Management System successfully achieves the project objectives and provides an effective solution for financial management.

9.2 Future Enhancements

Although the proposed system provides efficient expense management features, several enhancements can be implemented in the future to improve functionality and user experience.

The future enhancements of the system include:

- Development of a mobile application version for Android and iOS.
- Integration with online banking and digital payment systems.

- Addition of AI-based expense prediction and financial suggestions.
- Cloud storage for accessing data from multiple devices.
- Advanced analytics and spending prediction reports.
- Notification system using SMS or email reminders.
- Multi-language support for better accessibility.
- Enhanced security features such as OTP verification and encrypted passwords.

These future improvements can make FinTrack: Family and Expense Management System more powerful, intelligent and accessible for users.

APPENDIX

Source Code

The following source code snippets represent the technologies and implementation used in FinTrack: Family and Expense Management System. The system is developed using Python Flask for backend processing, HTML for webpage structure and CSS for user interface styling.

```
@app.route("/login", methods=["POST"])
```

```
def login():
```

```
    user = User.query.filter_by(
        email=request.form["email"],
        password=request.form["password"]
    ).first()
```

```
    if user:
```

```
        session["user_id"] = user.id
```

```
        return redirect("/mode")
```

```
    return "Invalid Login"
```

```
@app.route("/logout")
```

```
def logout():
```

```
    session.clear()
```

```
    return redirect("/")
```



```
{% extends "base.html" %}
```

```
{% block content %}
```

```
<h2>Family Setup</h2>
```

```
<h3>Create Family</h3>
```

```
<form action="/create_family" method="post">
```

```
<input name="name" placeholder="Family Name">
```

```
<button>Create</button>
```

```
</form>
```

```
<h3>Join Family</h3>
```

```
<form action="/join_family" method="post">
```

```
<input name="family_id" placeholder="Enter Family ID">
```

```
<button>Join</button>
```

```
</form>
```

```
{% endblock %}
```

Sample Outputs

The following outputs demonstrate the successful execution of the system and generated financial reports.

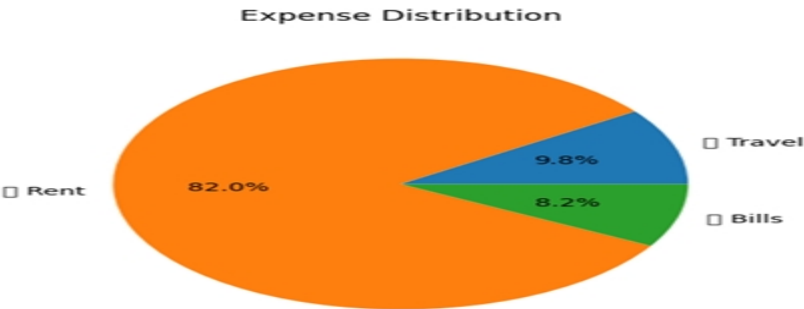
Smart Expense Tracker Report

Personal Expense Report

Total Expense: ■6100.0

Category	Amount	Date	Payment
■ Travel	600.0	2026-05-01	UPI
■ Rent	5000.0	2026-02-05	Debit Card
■ Bills	500.0	2026-03-12	Cash

Expense Distribution



Suggestion: Highest spending is on ■ Rent. Try reducing it.

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